USB-C

J1
USB-C (USB4105)
VBUS
CC1
CC2
D−
D−
D+
D+
SHIELD
GND
SBU1
SBU2
A9
B9
A5
B5
A7
B7
A6
B6
A8
B8
USB_DM
USB_DP
SH
GND
SMF5.0A
VBUS_F
F1
1A fast (466)
D4
SS14
+5V
R14
5.1k
R13
5.1k
D5
TPD2S017
CH1_OUT
GND
CH1_IN
CH2_OUT
VCC
CH2_IN
USB_DM_ESP
USB_DP_ESP
VBUS_F

POWER

U2
SGM2212-3.3
VIN
VOUT
GND
C2
10uF
C4
10uF
+5V
+3V3

ESP32-C6 MCU

ESP32-C6-WROOM-1-N8
ESP32-C6-WROOM-1
EN/CHIP_PU
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
MTMS/GPIO4/ADC1_CH4
MTDI/GPIO5/ADC1_CH5
MTCK/GPIO6/ADC1_CH6 U1
MTDO/GPIO7
NC
3V3
U0TXD/GPIO16
U0RXD/GPIO17
GPIO13/USB_D+
GPIO12/USB_D−
GPIO8
GPIO9
GPIO10
GPIO11
GPIO15
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
I2S_WS
OC3_OUT
OC2_OUT
I2S_MCLK
I2S_DOUT
USB_DP_ESP
USB_DM_ESP
BOOT
I2S_DIN
I2S_BCLK
I2C_SDA
I2C_SCL
GATE2_DRV
GATE3_DRV
GATE1_PRE
C3
10uF
C6
100nF
R12
10k
R6
100

POWER LED

D6
PWR
R15
1k

RELAY DRIVERS

Q1
2N7002
D1
1N4148W
K1
G6K-2F-Y 4.5V
R9
10k
GATE1
IN_P4
P2
Q2
2N7002
D2
1N4148W
K2
G6K-2F-Y 4.5V
R4
100
R7
10k
GATE2_DRV
R16
2.2k
R3
Q3
2N7002
D3
1N4148W
K3
G6K-2F-Y 4.5V
R5
100
R8
10k
GATE3_DRV
P4
GATE1_PRE

RESET / BOOT

R10
10k
C5
1uF
SW2
EN
R11
10k
SW1
BOOT

BELL SENSE

SW3
SW4
SW5
Polarity
P1
P2
P3
P4
P5
IN_P4
D7
D8
D9
1N4148W
OK1
OK2
OK3
LTV-217 (PC817)
OC1_OUT
OC2_OUT
OC3_OUT
OC1_IP
OC2_IP
OC3_IP
R17
5.1k
R1
5.1k
R2
5.1k
R3
1k
WF26 (6-way screw)
Pin_1
Pin_2
Pin_3
Pin_4
Pin_5
Pin_6

AUDIO ES8311 + LINE XFMR

U3
ES8311
CCLK
MCLK
PVDD
DVDD
DGND
SCLK/DMIC_SCL
ASDOUT
LRCK
DSDIN
AGND
EP
CE
CDATA
MIC1P/DMIC_SDA
MIC1N
VMID
ADCVREF
DACVREF
OUTN
OUTP
AVDD
I2C_SCL
I2C_SDA
I2S_MCLK
I2S_BCLK
I2S_DIN
I2S_WS
I2S_DOUT
R19
10k
R18
10k
R20
10k
C7
100nF
C8
100nF
C9
100nF
C10
10uF
C11
10uF
C12
1uF
C13
1uF
C14
1uF
C15
1uF
C17
1uF
C16
1uF
T1
SM-LP-5001

Sheet: /
File: doorbell.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 1/1